

Reliable and Energy-Efficient Communications via Collaborative Beamforming for UAV Networks

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Abstract—Unmanned aerial vehicles (UAVs) have been demonstrated to be a prominent component for wireless communications. In this work, we consider an emergency communication scenario wherein a UAV-based relay system collects data from ground users, and then uses different UAV-enabled virtual antenna arrays (UVAAs) to transmit the collected data to several remote base stations (BSs) via collaborative beamforming (CB). However, several adjacent aerial users (AUs) are carrying out other missions at the same time, which may be interfered by the signal transmitted by the UVAAs. Thus, we formulate a reliable and energy-efficient communication multi-objective optimization problem (RECMOP) to jointly maximize the minimum receiving signal-to-noise ratio (SNR) of the BSs, minimize the maximum average receiving SNR of the AUs, and minimize the propulsion power consumption of the UAVs, so that diminishing the energy cost while enhancing the system performance. The formulated

RECMOP is intricate since it is proven to be NP-hard and non-convex. Therefore, an improved multi-objective gravitational search algorithm (IMOGSA) with several specific designs is proposed to handle the formulated problem. Simulation results manifest that the proposed IMOGSA can effectively solve the formulated RECMOP, and it outperforms other benchmarks in both smaller and larger scale UAV networks. Moreover, extended simulation demonstrates the robustness of the proposed CB-based approach under several unexpected circumstances.

Index Terms—UAVs, virtual antenna array, collaborative beamforming, multi-objective optimization, reliable communication.

I. INTRODUCTION

EMERGENCY communication network is urgent to be established when the terrestrial infrastructure malfunctions, especially for the hard to reach areas [2], [3]. Due to the inherent characteristics of agile mobility, cost-effective, maneuverability and flexible deployment, the enthusiasm for dispatching unmanned aerial vehicles (UAVs) in emergency communications has skyrocketed since UAVs can provide reliable and efficient wireless communication services [4], [5], [6]. In conventional multiple UAVs-assisted cooperative transmission for emergency communications, UAVs are individually partitioned, with each UAV independently transmitting its own data. However, this mechanism has certain limitations, i.e., the UAVs are unable to communicate with the remote users due to the limited onboard resources of a UAV.

Collaborative beamforming (CB) has been demonstrated to be a feasible method in addressing the problems above [7]. To be specific, several UAVs can form a UAV-enabled virtual antenna array (UVAA) and adopt CB to transmit data to the legitimate base station (BS) directly. The advantages of applying CB are presented as follows. *First*, an N^2 fold gain in the received power at the receiver will be induced via N array elements. Accordingly, a beampattern with a high-gain main lobe pointing at the target receiver will be formed, then the long-distance transmission can be achieved. Moreover, due to the high gain, the data transmission task between UAVs and the receiver can be completed in a shorter time [8], [9]. *Second*, the participating UAVs are merely required to slightly adjust respective excitation current weight and placement, which will enhance the energy efficiency of UAVs. *However, there still has a major issue that need to be tackled when applying CB in UAV networks.* That is, the sidelobe levels (SLLs)

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of the UVAA system will interfere with the adjacent aerial users (AUs). Specifically, in some scenarios, multiple UAVs are deployed in the sky and the central controller chooses the appropriate UAVs to execute missions [10]. In this case, the other unselected UAVs maintain hovering to save energy and wait for other commands from the controller, which will inevitably receive the signal that is transmitted by the selected UAVs. In other words, the unselected UAVs will be interfered.

Performing an appropriate UVAA can obtain a beam pattern with a high-gain mainlobe and inferior low-gain sidelobes, which will significantly improve the signal strength to the legitimate BSs and reduce the interference caused to the AUs, respectively. However, there are several factors that will influence performance of the considered UVAA system. Specifically, the UAVs can move to appropriate locations to improve the directivity of the mainlobe and suppress the SLLs. Moreover, the excitation current weight of each participating UAV is a critical factor that will impact the performance of the UVAA system. Thus, how to select the suitable locations and excitation current weights of UAVs is a critical issue.

In this paper, we apply CB to achieve reliable and energy-efficient communications for UAV networks. Accordingly, a multi-objective optimization problem is formulated to enhance the communication performance of the UVAA system and reduce the propulsion power consumption of all the participating UAVs. The main contributions of this work are listed as follows:

- **CB-based UVAA System and Multi-objective Optimization Problem Formulation:** We consider an emergency communication scenario where a UAV-based relay system harvests data from ground users and then performs different UVAAs to send the data to several BSs via CB. At the same time, several AUs that are carrying out other assignments may be interfered by the transmission process between the UVAA system and BSs. Then, we formulate a reliable and energy-efficient communication multi-objective optimization problem (RECMOP) to maximize the minimum receiving signal-to-noise ratio (SNR) of the BSs, minimize the maximum average receiving SNR of the AUs, and minimize the propulsion power consumption of all the UAVs.
- **Meta-heuristic Algorithm:** We propose an improved multi-objective gravitational search algorithm (IMOGSA) with several specific designs to solve the formulated RECMOP. Specifically, IMOGSA adopts quasi-opposition based learning (QBL) strategy and discrete solution update strategy to enhance the quality of the initial solutions and deal with the discrete dimensions of solutions, respectively. Moreover, the proposed algorithm introduces an archive optimization method to improve the quality of solutions in the archive.
- **Performance Analysis:** Simulation results validate the effectiveness of the proposed IMOGSA in dealing with the formulated RECMOP. Moreover importantly, the proposed IMOGSA outperforms other benchmarks in both smaller and larger scale UAV networks. In addition, the performance analysis of the UVAA system under several unexpected circumstances is conducted, and then

the robustness of the proposed CB-based approach is shown.

The rest structure of this paper is organized as follows. Section II reviews several related works. Section III presents the models and preliminaries. Section IV formulates the RECMOP. Section V proposes the algorithm. Section VI shows the simulation results and Section VII concludes this work.

II. RELATED WORK

In this paper, we consider to achieve reliable and energy-efficient communications of the UVAA system via using CB, and several related works are presented as follows. Moreover, the main comparisons between related works and this work are presented in Table I, and the details are discussed as follows.

A. UAV-Assisted Emergency Communications

Some previous works have focused on deploying UAV to assist emergency communications. For example, Hu et al. [11] investigated the network recovery of disaster area based on the assistance of UAV and achieved the maximization of the uplink throughput through jointly optimizing the height of UAV, power control and bandwidth allocation. Huang et al. [12] considered the secure transmission in the emergency scenario and proposed a framework for UAV path designing to facilitate the data transmission and collection. Jiang et al. [13] studied the extension of life cycle with respect to medical-emergency facilities, and proposed a mechanism to deploy UAV to charge the wearable devices wirelessly. Lin et al. [14] considered to deploy a UAV as an aerial BS and attempted to seek out the optimal path to serve as many users as possible, subject to the limited battery capacity. Feng et al. [15] deployed UAVs to achieve wireless power transfer for Internet of Things (IoT) devices in emergency communication scenario. Zhang et al. [16] constructed a UAV-enabled emergency communication network wherein a UAV is deployed as mobile BS to achieve information collection from ground users. Moreover, Shah et al. [17] investigated UAV-assisted mobile edge computing to assist the emergency communication in disaster areas.

However, these aforementioned works concentrated on the improvement of communication performance, while none of them considered the energy consumption. Moreover, most of them only considered to dispatch a single UAV to assist the post-disaster relief, which is contrary to the practical scenarios.

B. UAV-Based Relay System

There are several works that deploy the UAV-based relay system to facilitate the wireless communications between the source node and destination node. For example, Zeng et al. [18] deployed the UAV as a mobile relay to assist the wireless communication, and studied the throughput maximization problem under the mobility constraint. Zhang et al. [19] regarded the UAV as an amplify-and-forward relay and aimed to minimize the outage probability of

TABLE I
COMPARISONS BETWEEN RELATED WORKS WITH THIS WORK

Reference	Considered scenario	Relay strategies	Optimization objective		Optimization variables		Method
	Emergency communications	Multiple UAVs-based relay communications	SNR	Energy	UAV location	Excitation current weight	GSA, QBL strategy, Discrete solution update strategy, Archive optimization method
[11]	✓	×	×	×	✓	×	×
[12]	✓	×	×	×	✓	×	×
[13]	✓	×	×	×	✓	×	×
[16]	✓	×	×	×	✓	×	×
[24]	×	✓	×	×	×	×	×
[25]	✓	✓	×	×	×	×	×
[27]	×	×	✓	×	✓	×	×
[28]	×	×	✓	×	✓	×	×
[29]	×	×	✓	×	×	×	×
[31]	×	✓	✓	×	×	×	×
[32]	×	×	×	✓	×	×	×
[34]	×	×	×	✓	×	×	×
[35]	×	×	×	✓	×	×	×
[38]	×	×	×	✓	✓	×	×
[40]	×	×	×	×	✓	×	×
This work	✓	✓	✓	✓	✓	✓	✓

the relay node through designing the trajectory and transmit power of UAV. Wang et al. [20] investigated the secrecy rate maximization problem in the presence of an eavesdropper, and developed an iterative algorithm to handle the aforementioned problem. Wang et al. [21] deployed a UAV to forward the transmitted data from the cell-edge users towards the cellular BS. Wei et al. [22] integrated the intelligent reflecting surface (IRS) with UAV to assist the communications between the access point and multiple users. Su et al. [23] considered an IRS-UAV architecture to connect the BS and the user.

However, these aforementioned works considered the single UAV-based relay systems, which implies that they are incapable of handling the long-transmissions since the transmit power of a single UAV is limited.

Several works investigated the multiple UAVs-enabled relay systems. For example, Khallaf et al. [24] dispatched UAVs to establish a free-space optical communication link between a source and a destination. Yang et al. [25] considered a disaster scenario and deployed a UAV-based ad hoc networks over the affected area to provide communication services. Moreover, Yin et al. [26] proposed a planning scheme for relay UAVs to enable the data transmission between UAVs and ground device.

However, the locations of UAVs are scattered in the above-mentioned works, which means that the completion time of the task will be inevitably extended.

C. UAV Communications Optimizations

There are several works which regard the SNR or the propulsion power consumption as an effective metric to evaluate the performance of a communication network.

Some existing works have considered to maximize the SNR of the system. For example, Lu et al. [27] proposed to adopt aerial intelligent reflecting surface (AIRS) which has high deployment flexibility and wider reflection of signal to assist wireless networks, and they maximized the worst-case SNR through obtaining the optimal beamforming of the

source node, the deployment and beamforming of the AIRS. Sabzehali et al. [28] considered a mmWave-based UAV communication scenario, and investigated the SNR maximization through determining the optimal 3D deployment and the orientation of UAVs. Park et al. [29] formulated an optimization problem to maximize the lower bound of the SNR in the downlink distributed antenna systems. Bushnaq et al. [30] studied a tethered UAV and regular UAV-assisted cellular traffic offloading scenario, and proposed a user association mechanism to maximize the end-to-end SNR. Then, they obtained the SNR coverage probability which refers to the probability that the received SNR is larger than a given threshold. Moreover, Sharma et al. [31] investigated a hybrid satellite-terrestrial network in which several UAV relays assist the wireless communications, and they evaluated the performance of SNR outage probability based on appropriate UAV relay selection.

Some existing works have considered to optimize the propulsion power consumption. For example, Cai et al. [32] maximized the energy efficiency by simultaneously optimizing the velocity, user scheduling, trajectory and transmit power. Sambo et al. [33] designed the UAV trajectory for the sake of minimizing the energy consumption. Moreover, Zeng et al. [34] derived the propulsion power consumption model firstly, and then achieved the minimization of propulsion power consumption through cooperatively designing the UAV trajectory, the communication time and the completion time of the whole task.

However, the aforementioned works only focused on single-objective optimization. In practical scenarios, multiple optimization objectives need to be jointly considered, i.e., the SNR and propulsion power consumption of UAVs are required to be simultaneously optimized.

D. Schemes for Solving Optimization Problems

Various schemes are used to handle the optimization problems in the domain of UAV communications and networks.

For example, Zhang et al. [35] deployed a UAV to execute data collection task for IoT devices, aiming to minimize the age of information (AoI) and energy cost. Then, they adopted weighted factors to transform the multi-objective optimization problem into a single-objective optimization problem. However, it is challenging to find an applicable mechanism for weight assignment since the objectives are conflicted with each other. Deng et al. [36] investigated the UAV-enabled covert communications, and adopted convex optimization to deal with the formulated problem to improve the covertness of the system. However, convex optimization needs to transform the optimization problem. Dai et al. [37] adopted deep reinforcement learning (DRL) to joint design the user association and trajectory in the multi-UAV networks. However, DRL takes a lot of time to train the model.

Moreover, the meta-heuristic algorithms are also widely used in tackling optimization problem. For example, Liang et al. [38] deployed UAVs to execute wireless power transfer, and formulated an optimization problem to enhance the charging efficiency and diminish the energy consumption of UAVs via optimizing the placement of UAVs. Then, they proposed an improved firefly algorithm to deal with the formulated problem. Roberge et al. [39] used the genetic algorithm and particle swarm optimization to design the flight path of UAV, given the characteristics of the UAV and the complexity of the environment. Moreover, Pham et al. [40] proposed to adopt the Harris hawks optimization algorithm to maximize the sum rate of ground users.

However, the abovementioned meta-heuristic algorithms are proposed to deal with the problems whose solutions only contain continuous dimensions, which means that they are unable to deal with the problems with mixed decision variables.

Accordingly, a multi-objective optimization problem is formulated to improve the performance of the communication network. Then, a meta-heuristic algorithm is proposed to handle the formulated problem.

III. MODELS AND PRELIMINARIES

A. System Model

As shown in Fig. 1, we consider an emergency communication scenario, where a UAV-based relay system consists several rotary-wing UAVs denoted as $\mathcal{U} = \{1, 2, \dots, N_U\}$ needs to harvest data from ground users in a monitor area, and then transmits the collected data to several legitimate BSs denoted as $\mathcal{B} = \{1, 2, \dots, N_{BS}\}$. During the transmission process above, several neighbouring AUs denoted as $\mathcal{AU} = \{1, 2, \dots, N_{AU}\}$ are performing other assignments, which will be interfered by the signal transmitted by the UAVs. The AUs can be detected by the UAVs equipped with cameras or radars. Moreover, the locations of the BSs and AUs are defined to be fixed and known.

In the considered network, the UAV-based relay system will cooperatively perform different UVAs to transmit data to several BSs by adopting CB. Mathematically, a three-dimensional (3D) Cartesian coordinate system is adopted wherein the locations of the i th UAV, the j th BS, and the k th AU are denoted by (x_i^U, y_i^U, z_i^U) , $(x_j^{BS}, y_j^{BS}, 0)$ and $(x_k^{AU}, y_k^{AU}, z_k^{AU})$, respectively.

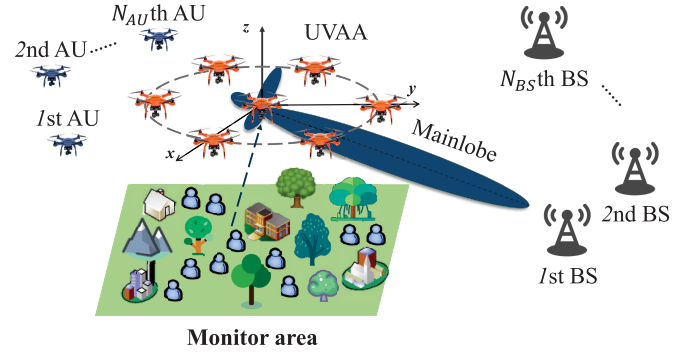


Fig. 1. Sketch map of a UVAA system in emergency communication scenario.

Remark 1: Once upon the collected data of a UAV reaches the upper limit or a UAV generates emergency data, the UAV needs to transmit the collected data to different remote BSs. Note that in certain scenarios, the BSs are not capable of establishing stable communication links with each other due to the long distance or other factors. Moreover, the collected data may need to be transmitted to different BSs for data backup or consistency check. Therefore, the UAVs are required to send the collected data to all the different remote BSs. $\square$

1) *Array Factor of the UVAA System:* The array factor (AF) for characterizing the beam pattern of the UVAA system can be expressed as follows [41]:

$$F(\theta, \phi) = \sum_{i=1}^{N_U} I_i e^{i_u [k_c (x_i^U \sin \theta \cos \phi + y_i^U \sin \theta \sin \phi + z_i^U \cos \theta)]}, \quad (1)$$

where I_i is the excitation current weight of the i th UAV, i_u represents the imaginary unit, $k_c = 2\pi/\lambda$ refers to the phase constant, and λ is the wavelength. Moreover, $\theta \in [0, \pi]$ and $\phi \in [-\pi, \pi]$ are the elevation and azimuth angles, respectively.

2) *Channel Model:* Two channel models are considered in this work. Specifically, the communication between the UVAA system and AUs belongs to air-to-air (A2A) communication, which can be the line-of-sight (LoS) model. Moreover, to model the channel between the UVAA system and BSs which can be regarded as air-to-ground (A2G) propagation channel, the probabilistic LoS and non-LoS (NLoS) models are required to be considered simultaneously.

The path loss between the UVAA system u and the receiver r is given as follows [42]:

$$L_{u,r} = \begin{cases} (K_o d_{u,r})^\alpha \mu_{LoS}, & \text{LoS link,} \\ (K_o d_{u,r})^\alpha \mu_{NLoS}, & \text{NLoS link,} \end{cases} \quad (2)$$

where $K_o = 4\pi f/c$, f denotes the carrier frequency and c is the speed of light, $d_{u,r}$ is the distance between the center of the UVAA system u and the receiver r . Note that the receiver r can be an AU or a remote BS. Moreover, α represents the path loss exponent, μ_{LoS} and μ_{NLoS} are the attenuation factors of LoS and NLoS communication links, respectively.

Accordingly, the path loss of the A2A channel from the UVAA system u to the k th AU is given as follows [42]:

$$L_{u,k}^{A2A} = (K_o d_{u,k})^\alpha \mu_{LoS}. \quad (3)$$

As for the A2G channel, the probability of having the LoS environment when the UVAA system u transmits data to the j th BS is as follows [43]:

$$P_{u,j}^{LoS}(\theta_{u,j}) = \frac{1}{1 + a \exp(-b(\theta_{u,j} - a))}, \quad (4)$$

where a and b are the modeling parameters, $\theta_{u,j}$ represents the elevation angle between the UVAA system u and the j th BS, which is gained by $\theta_{u,j} = \frac{180}{\pi} \sin^{-1}(\frac{\Delta z_{u,j}}{d_{u,j}})$, wherein $\Delta z_{u,j}$ and $d_{u,j}$ denote the vertical and horizontal distance between the UVAA system u and the j th receiver BS, respectively. Based on this, the probability of having the NLoS environment between the UVAA system u and the j th receiver BS can be described as follows:

$$P_{u,j}^{NLoS}(\theta_{u,j}) = 1 - P_{u,j}^{LoS}(\theta_{u,j}). \quad (5)$$

Accordingly, the average path loss of the A2G channel from the UVAA system u to the j th BS is given as follows [43]:

$$L_{u,j}^{A2G} = (K_o d_{u,j})^\alpha [\mu_{LoS} P_{u,j}^{LoS} + \mu_{NLoS} P_{u,j}^{NLoS}]. \quad (6)$$

3) *SNR*: SNR is a metric which can measure the signal strength and communication reliability by comparing the level of the desired signal to the level of noise. Specifically, the SNR from the UVAA system u to the j th BS, $SNR_{u,j}$, and that from the UVAA system u to the k th AU, $SNR_{u,k}$, are calculated as follows [44]:

$$SNR_{u,j} = 20 \log_{10} \left(\frac{P_u G_{u,j}}{L_{u,j}^{A2G} \sigma^2} \right), \quad (7)$$

and

$$SNR_{u,k} = 20 \log_{10} \left(\frac{P_u G_{u,k}}{L_{u,k}^{A2A} \sigma^2} \right), \quad (8)$$

where $P_u = \sum_{i=1}^{N_U} I_i^2 P_t$ is the transmit power of the whole UVAA system, wherein $I_i^2 P_t$ represents the actual transmit power of the i th UAV and each UAV element has the same maximum transmit power, i.e., P_t . Moreover, σ^2 refers to the noise power. In addition, $G_{u,j}$ and $G_{u,k}$ represent the gains of the UVAA system u to the j th BS and the k th AU, respectively, which are expressed as follows [45]:

$$G_{u,j} = \frac{4\pi |F(\theta_{u,j}, \phi_{u,j})|^2 w(\theta_{u,j}, \phi_{u,j})^2}{\int_0^{2\pi} \int_0^\pi |F(\theta, \phi)|^2 w(\theta, \phi)^2 \sin \theta d\theta d\phi} \eta, \quad (9)$$

and

$$G_{u,k} = \frac{4\pi |F(\theta_{u,k}, \phi_{u,k})|^2 w(\theta_{u,k}, \phi_{u,k})^2}{\int_0^{2\pi} \int_0^\pi |F(\theta, \phi)|^2 w(\theta, \phi)^2 \sin \theta d\theta d\phi} \eta, \quad (10)$$

where $(\theta_{u,j}, \phi_{u,j})$ and $(\theta_{u,k}, \phi_{u,k})$ are the directions of the j th BS and the k th AU, respectively. Moreover, $w(\theta, \phi)$ is the magnitude of the far-field beam pattern of each UAV element and $\eta \in [0, 1]$ is the antenna array efficiency.

B. Propulsion Power Consumption Model of UAV

For a rotary-wing UAV which flies in a 2D horizontal plane with speed v , the propulsion power consumption is given as follows [46]:

$$P(v) = P_B \left(1 + \frac{3v^2}{v_{tip}^2}\right) + P_I \left(\sqrt{1 + \frac{v^4}{4v_{iv}^4}} - \frac{v^2}{2v_{iv}^2}\right)^{\frac{1}{2}} + \frac{1}{2} d_0 \rho s A v^3, \quad (11)$$

where P_B and P_I are the blade profile and induced power in hovering status, respectively. Moreover, v_{tip} refers to the tip speed of the rotor blade, v_{iv} is the mean rotor induced velocity in hovering. In addition, d_0 , ρ , s and A denote the fuselage drag ratio, air density, rotor solidity and rotor disc area, respectively.

According to Eq. (11), the propulsion power consumption of a rotary-wing UAV flying in a 3D horizontal plane is presented as follows [46]:

$$E(T) \approx \int_0^T P(v(t)) dt + \frac{1}{2} m_D (v(T)^2 - v(0)^2) + m_D g (h(T) - h(0)), \quad (12)$$

where T refers to the end time of the flight, $v(t)$ denotes the instantaneous UAV speed at time t , m_D and g are the mass of UAV and gravitational acceleration, respectively. Moreover, $v(T)$ and $h(T)$ represent the speed and height of the UAV at time T , respectively. In addition, $v(0)$ and $h(0)$ are the speed and height of the UAV at the initial moment, respectively.

Remark 2: In this work, the trajectories of UAVs are not optimized, and each UAV will adopt a strategy of horizontal flight followed by vertical flight. The reasons are briefly listed as follows. *First*, due to the lack of an accurate 3D energy consumption model for rotary-wing UAVs at present, several existing works suggest to model the oblique flight of a UAV in a 3D space as the horizontal and vertical components [47], [48], [49]. *Second*, complex flights of UAVs may induce uncertainty to the autonomous flights of rotary-wing UAVs, thus the adopted flight strategy possesses the ability to stabilize the flight process. Note that the collision among UAVs will not occur since the minimum distance between two adjacent UAVs is added into the constraints of the following formulated optimization problem. $\square$

C. Multi-Objective Optimization

Multi-objective optimization problem (MOP) involves more than one objective function that should be optimized simultaneously. Mathematically, an MOP can be formulated as follows [50]:

$$\min_X F = [f_1(x), f_2(x), \dots, f_{N_o}(x)], \quad (13)$$

where x is the decision variable, X is a vector which contains all feasible decision variables. Moreover, N_o is the number of objective functions, and $f_o(x)$ represents the o th objective function.

Considering the conflicting relationship among the multiple objectives, the definition of Pareto dominance is introduced.

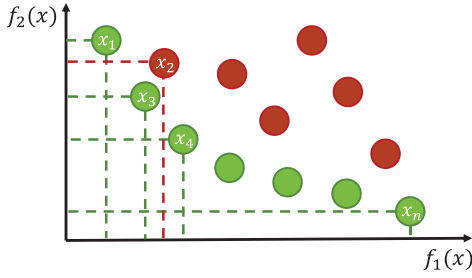


Fig. 2. Solution distribution of a bi-objective optimization problem.

To be specific, x_1 is defined to dominate x_2 if and only if $\forall o \in N_o, f_o(x_1) \leq f_o(x_2)$ and $\exists o \in N_o, f_o(x_1) < f_o(x_2)$. Accordingly, the solution which is not dominated by any other solutions is called Pareto optimal solution. Through taking the bi-objective minimization optimization problem as an example, Fig. 2 demonstrates the domination relationship among solutions in a more intuitive way. As can be seen, x_3 dominates x_2 . Moreover, $X^* = \{x_1, x_3, x_4, \dots, x_n\}$ is a set of Pareto optimal solutions.

IV. PROBLEM FORMULATION AND ANALYSIS

In this section, the optimization problem is formulated, and then the relevant analysis of the problem is given.

A. Problem Formulation

The optimization problem contains three optimization objectives. The first objective is to maximize the minimum SNR of all the BSs, which can be achieved by adjusting the locations as well as excitation current weights of the participating UAVs and controlling the order that the UVAA system transmits data to the BSs. To control interference caused by the transmission process on the AUs, the second objective is introduced, which is to minimize the maximum average receiving SNR of the AUs. Moreover, shifting locations of UAVs will consume more propulsion power, and thus we define minimizing the propulsion power consumption of UAVs as the third optimization objective.

We define $X = [\mathbb{C}^{\mathcal{U}}, \mathbb{I}^{\mathcal{U}}, \mathbb{O}]$ as the solution of the formulated RECMOP. To be specific, $\mathbb{C}^{\mathcal{U}} = \{\mathbb{C}_1^{\mathcal{U}}, \mathbb{C}_2^{\mathcal{U}}, \dots, \mathbb{C}_{N_U}^{\mathcal{U}}\}$ is the coordinate matrix of the UVAA system with $\mathbb{C}_i^{\mathcal{U}} = \{x_i^U, y_i^U, z_i^U\}$ denotes the location of the i th UAV. Moreover, $\mathbb{I}^{\mathcal{U}} = \{I_i | \forall i \in \mathcal{U}\}$ represents the set of excitation current weights of UAVs, and $\mathbb{O}$ denotes the order that the UVAA system transmits data to different BSs. Accordingly, the optimization objectives are described as follows.

1) *Optimization Objective 1*: The first objective is to maximize the minimum receiving SNR of the several remote BSs so that the quality of signal received by each BS can be guaranteed. Accordingly, the first objective function is designed as follows:

$$f_1(\mathbb{C}^{\mathcal{U}}, \mathbb{I}^{\mathcal{U}}, \mathbb{O}) = \widetilde{\text{Min}}_{j \in \{1, 2, \dots, N_{BS}\}} \{SNR_{u,j}\}, \quad (14)$$

where $\widetilde{\text{Min}}(\cdot)$ is the operator that calculates the minimum value of a vector.

Remark 3: In this work, we choose to optimize SNR instead of transmission rate and the reasons are as follows. *First*, SNR is a paramount metric which is capable of evaluating the communication performance of a network, and it puts more emphasis on the quality of the signal. For instance, in certain scenarios, the communication link is defined as interruption if the receiving SNR is lower than the predefined threshold [51], which means that the data transmission process between the transmitter and the receiver is no longer meaningful. *Second*, according to the Shannon-Hartley theorem, a high transmission rate will be obtained if the SNR is comparatively high. In other words, transmission rate and SNR of certain communication network have the same change tendency [52]. Thus, the SNR is selected to be optimized in this work. $\square$

2) *Optimization Objective 2*: The positions of the several AUs are assumed to be known and stationary in the considered scenario, which is typical in a practical scenario, which implies that the interference caused by the UVAA system on the AUs can be mitigated by minimizing the maximum average receiving SNR of the AUs when the UVAA system transmits data to different BSs. Thus, the second objective function is expressed as follows:

$$f_2(\mathbb{C}^{\mathcal{U}}, \mathbb{I}^{\mathcal{U}}, \mathbb{O}) = \widetilde{\text{Max}}_{j \in \{1, 2, \dots, N_{BS}\}} \left\{ \sum_{k=1}^{N_{AU}} \frac{SNR_{u,k}^j}{N_{AU}} \right\}. \quad (15)$$

where $SNR_{u,k}^j$ denotes the receiving SNR of the k th AU when the UVAA system transmits data to the j th BS.

3) *Optimization Objective 3*: The UAVs will move to suitable locations to perform an optimal UVAA for transmitting data to each receiver BS. However, additional propulsion power consumption will be induced during the flight process. Therefore, the third optimization objective is to minimize the propulsion power consumption of UAVs, and the corresponding objective function is defined as follows:

$$f_3(\mathbb{C}^{\mathcal{U}}, \mathbb{O}) = \sum_{i=1}^{N_U} \sum_{j=1}^{N_{BS}} E_{i,j}(T_{i,j}), \quad (16)$$

where $T_{i,j}$ represents the duration for the i th UAV to perform a UVAA to transmit data to the j th BS, and $E_{i,j}$ represents the propulsion power consumption of the i th UAV for serving the j th BS.

Accordingly, the RECMOP is formulated as follows [53]:

$$\min_X F = \{-f_1, f_2, f_3\}, \quad (17a)$$

$$\text{s.t. } C1 : 0 \leq I_i \leq 1, \forall i \in \mathcal{U}, \quad (17b)$$

$$C2 : L_{min} \leq x_i^U \leq L_{max}, \forall i \in \mathcal{U}, \quad (17c)$$

$$C3 : L_{min} \leq y_i^U \leq L_{max}, \forall i \in \mathcal{U}, \quad (17d)$$

$$C4 : H_{min} \leq z_i^U \leq H_{max}, \forall i \in \mathcal{U}, \quad (17e)$$

$$C5 : V_{min} \leq v \leq V_{max}, \forall i \in \mathcal{U}, \quad (17f)$$

$$C6 : \mathbb{O}^{B \times 1} \in \mathcal{SO}, \quad (17g)$$

$$C7 : D_{i_1, i_2} \geq D_{min}, \forall i_1, i_2 \in \mathcal{U}, \quad (17h)$$

where the constraint (17b) confines the value of excitation current weight of the i th UAV I_i to $[0, 1]$, the constraints (17c) and (17d) indicate that the minimum and maximum ranges

when UAVs fly in the horizontal plane are L_{min} and L_{max} , respectively, and the constraint (17e) manifests that the minimum and maximum ranges when UAVs fly in the vertical plane are H_{min} and H_{max} , respectively. The constraint (17f) indicates the lower bound V_{min} and upper bound V_{max} of the velocity of the UAVs, and $\mathcal{SO}$ in the constraint (17g) is the set of orders that the UVAA system transmits data to N_{BS} BSs, which has $N_{BS}!$ permutations and is expressed as $\mathcal{SO} = \{\mathbb{O}_1^{\mathcal{B} \times 1}, \dots, \mathbb{O}_n^{\mathcal{B} \times 1}, \dots, \mathbb{O}_{N_{BS}!}^{\mathcal{B} \times 1}\}$, wherein $\mathbb{O}_n^{\mathcal{B} \times 1} = \{O_1, \dots, O_j, \dots, O_{N_{BS}} | 1 \leq j \leq N_{BS}, O_j \in \mathcal{B}\}$. For instance, $\mathbb{O}^{\mathcal{B} \times 1} = \{2, 5, \dots, 7\}$ denotes that the UVAA system will transmit data to the 2nd, 5th, ..., 7th BS sequentially. Moreover, the constraint (17h) indicates the minimum distance between two adjacent UAVs.

Remark 4: Note that the optimization directions of f_1 and f_2 are opposite. The reason is that we attempt to achieve the maximum of f_1 and the minimum of f_2 , so that the quality of the signal received by each BS will be improved and the interference on the AUs caused by the UVAA system can be reduced simultaneously. $\square$

B. Problem Analysis

Lemma 1: The first optimization objective function f_1 of the formulated RECMOP is NP-hard since it is a nonlinear multi-dimensional 0-1 knapsack problem.

Proof: We first consider that the locations of UAVs and the order of transmitting data to different BSs are fixed and known. Thus, the solution space of the first optimization objective function only contains the excitation current weights of UAVs and it is expressed as $X' = [I_1, I_2, \dots, I_{N_U}]$. In this case, the simplified first optimization objective function is defined as S-RECMOP- f_1 and it is expressed as follows:

$$\min_{X'} -f_1 = -\widetilde{\text{Min}}_{j \in \{1, 2, \dots, N_{BS}\}} \{SNR_{u,j}\}, \quad (18a)$$

$$\text{s.t.} \quad C1: I_i \in \{0, 1\}, \forall i \in \{1, 2, \dots, N_U\}, \quad (18b)$$

$$C2: \sum_{i=1}^{N_U} I_i < N_U. \quad (18c)$$

The S-RECMOP- f_1 is a nonlinear multi-dimensional 0-1 knapsack problem which has been proven to be NP-hard [54]. Thus, the S-RECMOP- f_1 is NP-hard. $\blacksquare$

Lemma 2: The second optimization objective function f_2 of the formulated RECMOP is NP-hard.

Proof: The proof is similar to that of Lemma 1, and thus it is omitted. $\blacksquare$

Lemma 3: The third optimization objective function f_3 of the formulated RECMOP is NP-hard since it is a typical traveling salesman problem.

Proof: To ease of presentation, the locations and excitation current weights of UAVs are regarded as the constants in this part. Then, the solution space can be expressed as $X'' = [\mathbb{O}]$. Accordingly, the simplified third optimization objective function is defined as S-RECMOP- f_3 , and it is reformulated as follows:

$$\min_{X''} f_3 = \sum_{j=1}^{N_{BS}-1} E_{\mathcal{O}_j, \mathcal{O}_{j+1}}, \quad (19a)$$

$$\text{s.t.} \quad \mathbb{O}^{\mathcal{B} \times 1} \in \mathcal{SO}, \quad (19b)$$

where $E_{\mathcal{O}_j, \mathcal{O}_{j+1}}$ represents the propulsion power consumption of UAVs from transmitting data to the $\mathcal{O}_j$ th BS to that of the $\mathcal{O}_{j+1}$ th BS. Clearly, the reformulated S-RECMOP- f_3 is a typical traveling salesman problem that has been demonstrated to be NP-hard [55]. Thus, the S-RECMOP- f_3 is NP-hard. $\blacksquare$

Proposition 1: The formulated RECMOP is NP-hard since the subproblems are NP-hard.

Proof: The formulated RECMOP is NP-hard since Lemmas 1, 2 and 3 show that the subproblems f_1 , f_2 and f_3 are NP-hard. $\blacksquare$

Proposition 2: The formulated RECMOP is a large-scale optimization problem since the number of solution dimensions grows linearly with the number of UAVs or BSs.

Proof: The solution space of the formulated RECMOP contains 3D coordinates of UAVs ($\mathbb{C}^U$), excitation current weights of UAVs ($\mathbb{I}^U$) as well as the order of transmitting data to the BSs ($\mathbb{O}$), which means that there are $(4 \times N_U \times N_{BS} + N_{BS})$ solution dimensions should be optimized. In particular, the scale of the formulated RECMOP will grow with the number of UAVs or BSs increases. Thus, the formulated RECMOP is a large-scale optimization problem [56]. $\blacksquare$

Proposition 3: The formulated RECMOP can be classified into a mixed integer programming problem, thus it is non-convex.

Proof: The solution space of the formulated RECMOP contains continuous dimensions ($\mathbb{C}^U$, $\mathbb{I}^U$) and discrete dimensions ($\mathbb{O}$), so that the formulated RECMOP is a typical mixed integer programming problem (MINLP). Since MINLP is non-convex, the formulated RECMOP is also non-convex [57]. $\blacksquare$

V. PROPOSED OPTIMIZATION METHOD

In this section, the motivation of applying meta-heuristic algorithm in solving the formulated problem is first presented. Then, the main framework of the conventional MOGSA is introduced. Finally, IMOGSA is proposed to tackle the formulated RECMOP.

A. Motivation

There are three widely adopted methods to tackle the multi-objective optimization problems, which are DRL, convex optimization and meta-heuristic algorithms. DRL takes a lot of time to train the model, and it is more appropriate to deal with the problems with continuous time slots [58]. However, the formulated RECMOP is a single slot problem, which means that unnecessary overhead will be induced if the DRL is adopted to handle it. Moreover, the convex optimization is not good at tackling the problems with intricate constraints, e.g., the formulated RECMOP. In addition, meta-heuristic algorithms are efficient methods in dealing with large-scale optimization problems which are NP-hard and non-convex. Thus, a heuristic-based algorithm is proposed to solve the formulated RECMOP. *On the one hand*, meta-heuristic algorithms do not require a significant amount of time and cost to train models compared with the DRL. *On the other hand*, meta-heuristic algorithms do not need to transform the formulated problem as in convex optimization, which is equivalent to preserving the original solution space.

Among the various meta-heuristic algorithms, GSA is selected as the basic framework to deal with the formulated RECMOP, and the motivations are presented as follows.

- **The Unique Update Strategy of Agent:** In GSA, the position of each agent is updated based on the counter-parts of all the other agents in current population, while the positions of agents in other meta-heuristic algorithms are mainly updated based on the global and/or individual optima. Thus, compared with other meta-heuristic algorithms, GSA has a lower probability of falling into local optimum in handling certain optimization problems due to the unique update strategy.
- **Strong Global Search Capabilities:** During the process of calculating gravity, the force on any agent is expressed by the randomly weighted sum of the gravity of other agents, which helps introduce the uncertainty during the search process, making the algorithm to explore the search space more flexibly.
- **Simple Principle, Few Parameters and Easy to Implement:** Compared with some meta-heuristic algorithms, e.g., squirrel search algorithm [59] and harris hawks optimization [60], GSA is characterized by simple principle and few parameters, which facilitates the comprehension and implementation. In practical scenarios, the UAVs are required to execute the algorithm, hence the GSA will exhibit a shorter execution time and require less hardware resources.

In the following, the conventional GSA is briefly introduced.

B. Conventional GSA

Enlightened by the Newton's laws of gravitation and kinematics, GSA is a typical meta-heuristic algorithm and has been demonstrated to be efficient in finding the optimal solutions of different optimization problems [61], [62]. Specifically, the solution update method of GSA is expressed as follows [61]:

$$X_{i,d}^{t+1} = X_{i,d}^t + V_{i,d}^{t+1}, \quad (20)$$

$$V_{i,d}^{t+1} = rand \times V_{i,d}^t + a_{i,d}^t, \quad (21)$$

where $X_{i,d}^{t+1}$ and $V_{i,d}^{t+1}$ are the position and velocity of the i th agent in the d th dimension at the $t+1$ th iteration, respectively. Moreover, $rand$ is a random number between 0 and 1, $a_{i,d}^t$ refers to the acceleration of the i th agent in the d th dimension at the t th iteration, and it is given as follows [61]:

$$a_{i,d}^t = \frac{\sum_{j=1, j \neq i}^{N_{pop}} rand \times F_{ij,d}^t}{M_{ii}^t}, \quad (22)$$

where M_{ii}^t denotes the inertial mass of the i th agent. Moreover, $\sum_{j=1, j \neq i}^{N_{pop}} rand \times F_{ij,d}^t$ is the total force of the other agents acting on the i th agent, wherein N_{pop} refers to the number of agents in the population and $F_{ij,d}^t$ is the force of the j th agent acting on the i th agent in the d th dimension, and it is written as follows:

$$F_{ij,d}^t = G^t \frac{M_{pi}^t \times M_{aj}^t}{R_{ij}^t + \varepsilon} (X_{j,d}^t - X_{i,d}^t), \quad (23)$$

where G^t is the gravitational constant at the t th iteration which will change as the number of iterations increases, R_{ij}^t denotes

the Euclidian distance between the i th and j th agents at the t th iteration, ε is a constant with a small value. Moreover, M_{pi}^t and M_{aj}^t are the passive gravitational mass of the i th agent and the active gravitational mass of the j th agent, respectively.

Several multi-objective versions of GSA have been proposed to handle the multi-objective optimization problems, which can be referred to [63] and [64]. The archive is adopted to save the Pareto optimal solutions in these algorithms. However, these algorithms are proposed to deal with the optimization problems only with continuous solution dimensions, which implies that they are incapable of handling the formulated RECMOP with hybrid decision variables. Thus, we propose IMOGSA to overcome the abovementioned challenge and improve the quality of solutions when dealing with the formulated RECMOP.

C. IMOGSA

In this section, IMOGSA is introduced to deal with the formulated RECMOP. Specifically, IMOGSA adopts three specific designs which are the QBL strategy, the discrete solution update strategy and the archive optimization method. The details of the designs are shown as follows, and the general structure of IMOGSA is presented in Algorithm 1. Note that the number of agents in the population and the number of maximum iterations are denoted as N_{pop} and t_{max} , respectively. Moreover, P_t , F_t and A_t are the population, fitness value set of the population and archive at the t th iteration, respectively. In addition, P_{merge} is the mixed population and F_{merge} refers to the corresponding fitness value set.

1) **QBL Strategy:** According to Proposition 2, the formulated RECMOP is a large-scale optimization problem, which implies that the algorithm may be trapped into local optimal easily when the quality of initial solutions is not high. Thus, to cope with the characteristic of the formulated RECMOP, we introduce a multi-step initialization method called QBL strategy. Specifically, the procedures of QBL are presented as follows [65].

First, the continuous dimensions of solutions in the initial population X^I are generated, and the process is as follows:

$$X_{i,d}^I = LB_d + rand * (UB_d - LB_d), \quad d = 1, 2, \dots, D_c, \quad (24)$$

where $X_{i,d}^I$ represents the d th dimension of the i th solution in X^I , LB_d and UB_d refer to the lower and upper bounds of the d th dimension, respectively. Moreover, D_c denotes the number of continuous dimensions.

Second, the opposite population X^O of the initial population X^I is generated through using some heuristic rules, and the rule adopted in this work is given as follows:

$$X_{i,d}^O = UB_d + LB_d - X_{i,d}^I, \quad d = 1, 2, \dots, D_c, \quad (25)$$

where $X_{i,d}^O$ is the opposite point of $X_{i,d}^I$.

Third, the strategy gains the quasi-opposite population X^Q . Specifically, the continuous dimension of solutions in X^Q is generated as follows:

$$X_{i,d}^Q = M_d + rand * (X_{i,d}^O - M_d), \quad d = 1, 2, \dots, D_c, \quad (26)$$

Algorithm 1 IMOCSA

Input: N_{pop} , t_{max} , P_0 , F_0 , A_0 , etc.
Output: $A_{t_{max}}$.

```

1  $P_0 \leftarrow \emptyset$ ,  $F_0 \leftarrow \emptyset$ ,  $A_0 \leftarrow \emptyset$ ;
2 Initialize  $P_0$  by using QBL strategy;
3 for  $i = 1$  to  $N_{pop}$  do
4   Count  $F_{0i}$  of  $X_i$  in  $P_0$ :  $F_{0i} = [f_{i,1}, f_{i,2}, f_{i,3}]$ ;
5    $F_0 \leftarrow F_0 \cup F_{0i}$ ;
6 end
7 Update  $A_0$  based on  $F_0$ ;
8 for  $t = 1$  to  $t_{max}$  do
9   Update  $G^t$ ;
10  Calculate mass of agents in  $P_{t-1}$  and  $A_{t-1}$ ;
11  Select  $Food$  from  $A_{t-1}$  based on roulette selection;
12  for  $i = 1$  to  $N_{pop}$  do
13    Count  $a_i^t$  of the  $i$ th agent based on Eq. (22);
14    Update the continuous dimensions ( $\mathbb{C}^U$ ,  $\mathbb{I}^U$ ) of  $X_i$ 
      in  $P_t$  based on Eq. (20);
15    Update the discrete dimensions ( $\mathbb{O}$ ) of  $X_i$  in  $P_t$ 
      based on Algorithm 2;
16  end
17  Generate  $P_{cm}$  based on Algorithm 3;
18   $P_{merge} \leftarrow P_t \cup P_{cm}$ ;
19  Calculate  $F_{merge}$  of  $P_{merge}$ ;
20  Update  $A_t$  based on  $F_{merge}$ ;
21 end
22 Return  $A_{t_{max}}$ .
```

where M_d is the middle point which is calculated by $LB_d + (UB_d - LB_d)/2$.

Fourth, the initial population X^I and quasi-opposite population X^Q are integrated into a population symbolized as X^M , which is described as follows:

$$X^M = X^I \cup X^Q. \quad (27)$$

Finally, the strategy reserves a subset of the population X^M according to the population size and the dominant relationship between agents.

Note that the discrete dimensions of each solution in the populations above are generated randomly.

2) **Discrete Solution Update Strategy**: Each solution of the formulated RECMOP contains both continuous ($\mathbb{C}^U$, $\mathbb{I}^U$) and discrete ($\mathbb{O}$) dimensions.

The continuous dimensions can be updated by the mechanism of GSA. However, the conventional meta-heuristic algorithms, e.g., GSA, cannot deal with the discrete dimensions since they are initially proposed to handle the problems whose solution only contains continuous dimensions. Therefore, the discrete solution update strategy is proposed in this work, which introduces the order-based crossover (OBX) and subtour exchange crossover (SEX), as show in Fig. 3. Accordingly, the details of the discrete solution update strategy are given in Algorithm 2.

The main steps of OBX for generating one of the offsprings are summarized as follows:

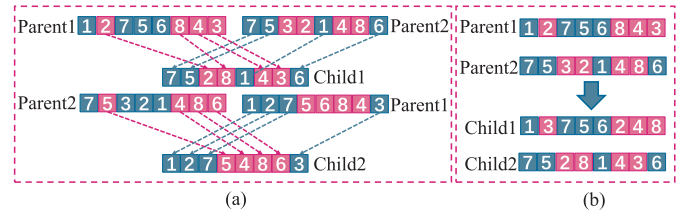


Fig. 3. Sketch maps of order-based crossover and subtour exchange crossover. (a) Order-based crossover. (b) Subtour exchange crossover.

Algorithm 2 Discrete Solution Update Strategy

```

1 Define the discrete dimensions of  $Food \mathbb{O}_{Food}$ .
2 if  $rand < 0.5$  then
3    $X_i(\mathbb{O})$  crosses with  $\mathbb{O}_{Food}$  based on OBX;
4 else
5    $X_i(\mathbb{O})$  crosses with  $\mathbb{O}_{Food}$  based on SEX;
6 end
7 Return  $X_i(\mathbb{O})$ ;
```

First, the OBX selects one sub-dimensions from *parent1* and confirms the positions of unselected elements in *parent2*.

Second, the method exploits the selected sub-dimensions of *parent1* and unselected elements of *parent2* to generate one of the offsprings.

The main procedures of SEX are depicted as follows:

First, one sub-dimensions is selected from *parent1* and the corresponding sequence of the sub-dimensions is ascertained in *parent2*.

Second, the SEX swaps the selected sub-dimensions while maintaining the positions of unselected elements unchanged.

3) **Archive Optimization Method**: Inspired by the non-dominated sorting genetic algorithm II (NSGA-II) [66], a new population is generated by using the crossover and mutation strategies on the solutions in the archive. Then, the new population is used to improve the quality of solutions in the archive.

The crossover operation of the continuous dimensions of two solutions X_{c1} and X_{c2} is expressed as follows:

$$\begin{cases} X'_{c1,d} = \beta * X_{c1,d} + (1 - \beta) * X_{c2,d} \\ X'_{c2,d} = \beta * X_{c2,d} + (1 - \beta) * X_{c1,d}, \end{cases} \quad (28)$$

where β is a random number matrix.

Moreover, the mutation operation concerning the continuous dimensions of solution X_m is designed as follows:

$$X'_{m,d} = X_{m,d} + \sigma * rand(1, n), \quad (29)$$

where σ represents the mutation step and $randn(1, n)$ denotes a random number matrix with 1 row and n column.

The crossover and mutation operations of the discrete dimensions are achieved by OBX and discrete mutation operator, respectively. Specifically, the discrete mutation operator is to select two different dimensions and exchange their positions.

The main procedures of the archive optimization method are presented in Algorithm 3. Note that N_c and N_m represent the

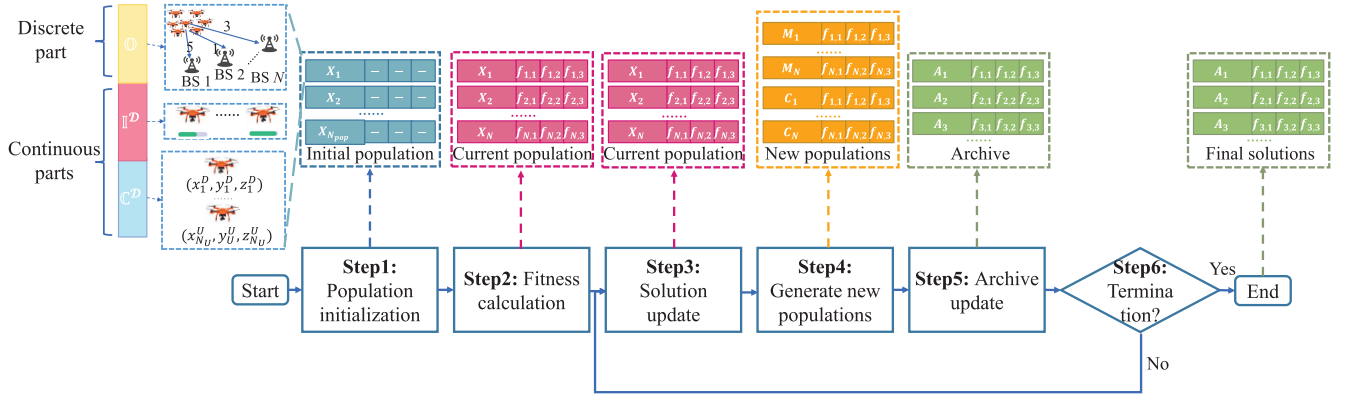


Fig. 4. The algorithm framework of IMOGSA for solving RECMOP.

Algorithm 3 Archive Optimization Method

Input: N_c, N_m, A_t .

Output: P_{cm} .

```

1  $P_c \leftarrow \emptyset, P_m \leftarrow \emptyset;$ 
2 for  $i = 1$  to  $N_c/2$  do
3   Select  $X_{c1}$  and  $X_{c2}$  from  $A_t$  randomly;
4   Generate  $X'_{c1}$  and  $X'_{c2}$  based on Eq. (28) and OBX;
5    $P_c \leftarrow P_c \cup X'_{c1} \cup X'_{c2};$ 
6 end
7 for  $i = 1$  to  $N_m$  do
8   Select  $X_m$  from  $A_t$ ;
9   Generate  $X'_m$  by mutating  $X_m$  based on Eq. (29) and
   discrete mutation operator;
10   $P_m \leftarrow P_m \cup X'_m;$ 
11 end
12  $P_{cm} \leftarrow P_c \cup P_m;$ 
13 Return  $P_{cm}$ .
```

numbers of solutions in crossover and mutation populations, respectively.

D. Solving RECMOP With IMOGSA

Fig. 4 shows the framework of the proposed IMOGSA for handling the formulated RECMOP, and the details are as follows.

- **Step 1:** The proposed IMOGSA generates the initial population. Specifically, the continuous dimensions ($\mathbb{C}^U, \mathbb{I}^U$) of each solution are initialized by using QBL strategy and the discrete dimensions ($\mathbb{O}$) are initialized by generating a random sequence.
- **Step 2:** The corresponding fitness value of each solution in the initial population is calculated and stored.
- **Step 3:** The solutions in current population are updated according to the solution update mechanism of the conventional GSA and the proposed discrete solution update strategy. Moreover, the fitness values of solutions in current population are calculated and updated accordingly.

- **Step 4:** The proposed IMOGSA generates two new populations, i.e., the crossover and the mutation populations, according to the introduced archive optimization method.
- **Step 5:** A set of dominant solutions are selected from the mixed population which contains current population and the two new populations. Then, these dominant solutions are saved in the archive.
- **Step 6:** If the termination condition is met, the IMOGSA will stop, and the solutions in the archive will be regarded as final solutions. Otherwise, the algorithm continues.

E. Analysis of the Proposed IMOGSA

In this part, the related analysis of the proposed IMOGSA is presented, which are the complexity analysis and convergence analysis.

Lemma 4: The computational complexity of the proposed IMOGSA is $\mathcal{O}(t_{max} \cdot N_o \cdot N_{pop}^2)$.

Proof: The number of objective functions, the population size and the maximum number of agents in the archive are symbolized as N_o , N_{pop} and N_{arc} , respectively. The computational complexity of the proposed IMOGSA is mainly decided by the calculation of objective functions and the rank of solutions in the archive. Accordingly, the computational complexity of computing objective functions is $\mathcal{O}(N_o \cdot N_{pop})$. Moreover, the computational complexity of ranking N_{arc} solutions of each objective function in each iteration is $\mathcal{O}(N_o \cdot N_{arc}^2)$. Based on the setting that N_{pop} is equal to N_{arc} , the computational complexity of ranking the solutions can be written as $\mathcal{O}(N_o \cdot N_{pop}^2)$. Hence, the overall computational complexity of the proposed IMOGSA is $\mathcal{O}(t_{max} \cdot N_o \cdot N_{pop}^2)$. ■

Proposition 4: The proposed IMOGSA will be converged during the iteration process.

Proof: Given that $V_i^t = X_i^t - X_i^{t-1}$, the solution update process of the i th agent at the t th iteration can be rewritten as follows:

$$X_i^{t+1} = X_i^t + V_i^{t+1}, \quad (30)$$

$$V_i^{t+1} = rand * (X_i^t - X_i^{t-1}) + a_i^t. \quad (31)$$

According to Eqs. (30) and (31), the solution update process can be expressed as follows:

$$X_i^{t+1} = (1 + rand) * X_i^t - rand * X_i^{t-1} + a_i^t. \quad (32)$$

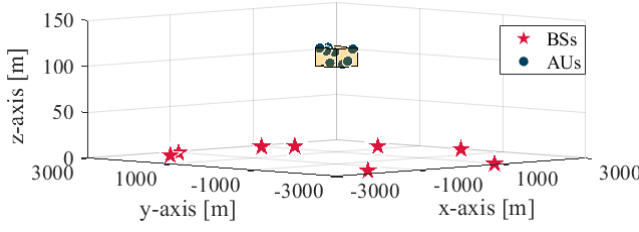


Fig. 5. The locations of BSs and AUs.

Then, the expectation of X_i^{t+1} can be described as follows:

$$E[X_i^{t+1}] = E[1 + rand] * E[X_i^t] - E[rand] * E[X_i^{t-1}] + E[a_i^t]. \quad (33)$$

By defining the lower and upper bounds of $rand$ as α and β , respectively, the recursive relationship among the positions of the i th agent at different iterations can be written as follows:

$$\begin{bmatrix} E[X_i^{t+1}] \\ E[X_i^t] \end{bmatrix} = \begin{bmatrix} \frac{2+\alpha+\beta}{2} & -\frac{\alpha+\beta}{2} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} E[X_i^t] \\ E[X_i^{t-1}] \end{bmatrix} + \begin{bmatrix} E[a_i^t] \\ 0 \end{bmatrix}. \quad (34)$$

For ease of presentation, the coefficient matrix is written as follows:

$$M_c = \begin{bmatrix} \frac{2+\alpha+\beta}{2} & -\frac{\alpha+\beta}{2} \\ 1 & 0 \end{bmatrix}. \quad (35)$$

According to [67] and [68], the necessary and sufficient condition to guarantee the convergence of the proposed IMOGSA is that the magnitude of the eigenvalues of M is smaller than 1. Given that α and β are set to be 0 and 1, respectively, we can calculate $|M_c| = (\alpha + \beta)/2 = 0.5 < 1$. Therefore, the proposed IMOGSA will converge within limited iterations. ■

VI. SIMULATION RESULTS

In this section, we conduct simulations to demonstrate the effectiveness of the proposed IMOGSA. First, the simulation setups are given. Second, the proposed IMOGSA is adopted to tackle the RECMOP, and the simulation results are compared with several other benchmarks. Finally, the performance of the UVAA system under several unexpected circumstances is analyzed and discussed.

A. Simulation Setups

To match reality, we consider smaller and larger scale UAV networks wherein the numbers of UAVs are set to be 8 and 16, respectively. In both of the considered networks, the UAVs are dispatched over a monitor area of $100 \times 100 \text{ m}^2$ for ease of control. To be specific, the minimum and maximum horizontal flight ranges of UAVs are set to be 0 m and 100 m, and the vertical flight range of UAVs is confined from 90 m to 120 m, to establish communication links with high LoS probability with the BSs, enabling faster transmission rate for the air-to-ground link. Moreover, the related settings of BSs follows the existing work [44]. Specifically, the number of BSs is set to be 8, the locations of BSs are shown in Fig. 5. Given that the coverage radius of a BS is typically around 1 to 5 km [69], thus

the considered BSs-related settings are reasonable. In addition, the locations of AUs are randomly generated near the UAVs. To be specific, the approximate area in which AUs are located is also shown in Fig. 5, wherein the horizontal and vertical locations of AUs are both confined from -200 m to 300 m. All the other simulation parameters follow the work in [70].

Several benchmark schemes are introduced for making comparisons with the proposed IMOGSA, which can be divided into two categories, i.e., meta-heuristic algorithms and the conventional multi-hop relay strategy, respectively.

- **Meta-heuristic algorithms:** Several meta-heuristic algorithms are introduced, which are NSGA-II [66], multi-objective dragonfly algorithm (MODA) [71], multi-objective grey wolf optimizer (MOGWO) [72], multi-objective salp swarm algorithm (MSSA) [73], and the conventional MOGSA. Note that the proposed discrete solution update strategy is adopted by the algorithms above so that they can deal with the discrete dimensions. Moreover, the number of maximum iterations and population size are defined to be 500 and 30, respectively.
- **Multi-hop relay strategy:** To make comparisons with the proposed CB-based approach, the conventional multi-hop UAV relay strategy is presented. To be specific, a group of UAVs are uniformly distributed between the monitor area and the receiver BSs successively.

B. Simulation Results

1) **Optimization Results Obtained by IMOGSA:** Table II presents the numerical results regarding the SNR of each BS and that of the adjacent AUs when the UVAA system transmits related data to the BS in both smaller and larger scale UAV networks. First, it is apparent that the receiving SNR of a BS in the larger scale UAV network is almost 2 to 3 times that of the BS in the smaller scale UAV network. The reason is that with the number of UAV elements increases, the beam will be sharper and the energy is more concentrated, resulting in a higher SNR of the direction of each BS. Second, it can be find that the SNR of AUs will also increase with the expansion of the scale of UAV network, due to the increased radiation intensity of sidelobes. Finally, it is observed that the receiving SNR of each remote BS is much lower than that of the adjacent AUs, which is induced by the long distance between each BS and the UVAA system. However, even in this case, the achievable rate of the BS is close to 3.16×10^6 bps given that the receiving SNR is 6 dB. In conclusion, the proposed IMOGSA can achieve satisfactory optimization results.

The moving paths of UAVs obtained by IMOGSA in both smaller and larger scale UAV networks for transmitting data to the first BS are visualized in Fig. 6. It is found that the optimized positions of UAVs are more compact and centralized than the initial positions of UAVs. The reason is that the centralized locations of elements contribute to a superior beam pattern with a high-gain mainlobe and low-gain sidelobes.

Ultimately, it is more preferable to deploy a larger scale UAV network if a higher receiving SNR of each BS is required to improve the communication performance. Moreover, the

TABLE II
THE OPTIMIZED SNR OF BSS AND AUs OBTAINED BY THE PROPOSED IMOCSA IN SMALLER AND LARGER SCALE UAV NETWORKS

SNR in smaller scale UAV network [dB]									SNR in larger scale UAV network [dB]								
BS			AUs						BS			AUs					
6.39	56.6	42.7	69.1	22.6	62.9	34.1	44.4	50.5	13.4	70.1	50.9	32.9	52.4	66.2	63.4	66.4	58.3
6.37	42.7	62.2	47.0	60.3	33.7	48.5	57.0	60.9	21.3	57.6	64.7	68.8	67.7	41.7	55.0	33.0	66.1
8.33	51.5	55.1	58.3	57.8	51.5	28.8	61.7	60.2	20.2	50.7	35.5	73.1	60.2	76.9	57.1	60.7	56.3
6.39	54.3	72.8	57.5	44.9	31.8	29.0	43.8	31.7	18.3	53.3	43.6	51.2	66.5	58.8	63.9	63.1	61.3
6.59	49.2	42.5	47.1	17.4	50.6	71.4	56.8	40.2	13.4	50.2	81.8	30.3	58.6	53.2	54.7	48.4	53.0
9.15	54.7	42.9	59.9	57.4	69.6	12.1	50.4	49.4	20.5	47.3	53.1	51.5	66.5	56.2	74.2	72.2	67.5
9.16	43.6	67.6	39.2	49.5	49.5	54.0	52.1	59.1	15.3	52.2	59.8	69.0	55.4	64.8	65.6	51.4	47.8
6.97	43.3	59.8	57.4	52.5	60.2	66.8	41.5	48.4	13.6	54.5	55.6	38.4	70.4	76.7	65.7	65.9	46.6

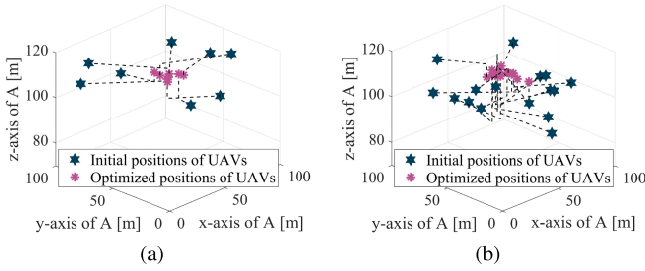


Fig. 6. Moving paths of UAVs obtained by the proposed IMOCSA. (a) Smaller scale UAV network. (b) Larger scale UAV network.

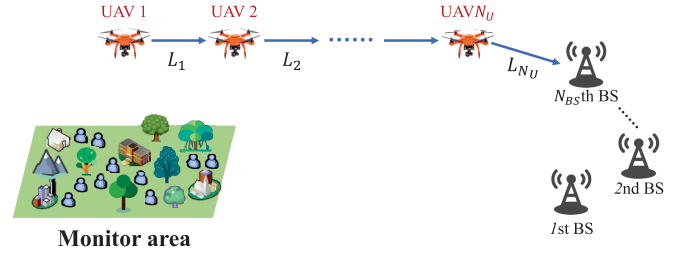


Fig. 8. Sketch map of the multi-hop UAV relay system.

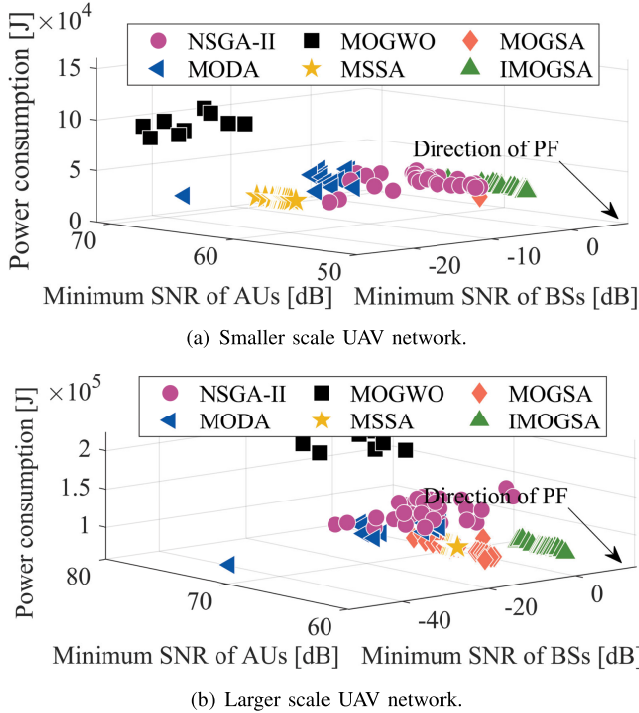


Fig. 7. Solution distributions obtained by the proposed IMOCSA and other meta-heuristic algorithms.

proposed CB-based approach is more suitable for smaller scale UAV network for reducing the propulsion power consumption of UAVs so that the UAVs can carry out more other missions.

2) *Comparison Results of IMOCSA and Other Meta-Heuristic Algorithms:* Fig. 7 illustrates the Pareto solution distributions obtained by the proposed IMOCSA and the other

TABLE III

COMPARISON RESULTS OF THE MULTI-HOP RELAY STRATEGY AND THE PROPOSED IMOCSA

Method	Smaller scale UAV network			Larger scale UAV network		
	f_1 [dB]	f_2 [dB]	f_3 [J]	f_1 [dB]	f_2 [dB]	f_3 [J]
Multi-hop	45.6	67.9	8.37×10^5	215.1	228.9	1.69×10^6
IMOCSA	6.3	57.6	2.11×10^4	13.4	65.8	5.99×10^4

meta-heuristic algorithms in both smaller and larger scale UAV networks. It is found that the solutions obtained by the proposed IMOCSA are closer to the direction of Pareto front (PF), which implies that the performance of the proposed IMOCSA is better than the other adopted meta-heuristic algorithms. The reason is that the introduction of QBL strategy and archive optimization method will enhance the quality of solutions.

3) *Comparison Results of IMOCSA and Multi-Hop UAV Relay Strategy:* In the constructed multi-hop UAV relay system, N_U UAVs at the same altitude are uniformly deployed between the center of the monitor area and a receiver BS, as shown in Fig. 8. Once upon the current transmission process is finished, each relay UAV will fly to the next pre-designed location to execute another transmission mission. Specifically, the decode-and-forward protocol is adopted, and the communication link between the i th and $i+1$ th relay UAVs is denoted as L_i^j when the receiver is the j th BS.

Table III presents the comparison results of the multi-hop relay strategy and the proposed IMOCSA. First, it is observed that compared with the proposed IMOCSA, the conventional multi-hop relay strategy will achieve exceptionally high receiving SNR of BSs. The reason is that the distance between the last-hop UAV and the receiver BS is comparatively close, which results in a low path loss and then the receiving SNR

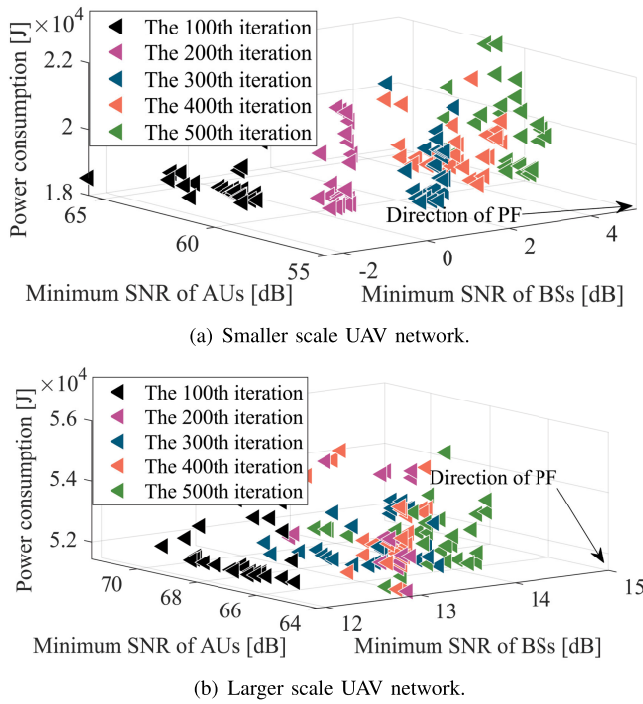


Fig. 9. Convergence analysis of the proposed IMOGSA. (a) Smaller scale UAV network. (b) Larger scale UAV network.

is improved accordingly. Similarly, the maximum average receiving SNR of AUs is also relatively high due to the close distance between the AUs and the UVAA system. Moreover, the multi-hop relay strategy will consume more propulsion power compared with IMOGSA since the UAVs are required to fly to respective remote position to construct an ad-hoc network. Generally speaking, if a very high receiving SNR or achievable rate of each BS is required, it is more appropriate to deploy a multi-hop relay network. Otherwise, the CB-based approach is more advisable.

4) *Convergence Analysis of the Proposed IMOGSA*: In this part, the convergence analysis of the proposed IMOGSA is conducted through analyzing the changing trend of the Pareto solutions with the number of iterations. Specifically, Fig. 9 presents the Pareto solutions obtained in the 100th, 200th, 300th, 400th, and 500th iterations. It can be observed from the figure that the Pareto solution set gradually approaches the direction of PF as the number of iterations increases in both smaller and larger scale UAV networks. Moreover, with the number of iterations increasing, the movement range of the Pareto solution set becomes smaller. Therefore, the proposed IMOGSA will be converged during the iteration process, which corresponds to Proposition 4.

C. Performance Analysis Under Unexpected Circumstances

In this section, several unexpected circumstances are taken into consideration, which are a single damaged UAV element, position jitters of UAVs, position jitters of AUs and imperfect phase synchronization, respectively. The performance of the proposed CB-based approach under the circumstances above is estimated. Note that the simulations are only carried out in smaller scale UAV network for simplicity.

1) *Single Damaged UAV*: In this part, an unexpected circumstance where the component of a UAV is damaged so that the UAV is not capable of participating in the communication process is considered. In this case, other UAVs that exist in the system will continue to carry out the CB mission. To better mimic the real scenarios, the damaged UAV in the whole UVAA system is selected randomly, and it is also undefined that how many BSs that the UAV has served before it is damaged. Note that we only consider the situation that the UAV is damaged in the moment when it is prepared to serve the next BS after serving the former BS, and the scenarios in which the UAV is damaged during the process of sending data to a specific BS will be left as our future work.

Fig. 10 shows the performance of the UVAA system where a UAV element is damaged. To be specific, Figs. 10(a) and 10(b) signify that the damaged UAV has a serious impact on the minimum receiving SNR of BSs and the maximum average receiving SNR of AUs. Moreover, it is reasonable that the propulsion power consumption of UAVs is reduced since the damaged UAV is no longer participating in the communication, as shown in Fig. 10(c). Overall, we find that the proposed CB-based approach still has comparatively decent performance even when a single UAV element is damaged.

2) *Position Jitters of UAVs*: Due to the influence of wind, airflow, rainstorm, etc., the positions of UAVs will be shifted, which may yield a non-negligible impact on the performance of the UVAA system. Therefore, we conduct simulations to demonstrate how the system will operate when the position jitters of UAVs occur. Specifically, the normal distribution is exploited to generate random position jitters, and the maximum jitters of UAVs in 3D directions are designed as 0.2 m, 0.4 m, 0.6 m, 0.8 m and 1 m, respectively [74].

Fig. 11 intuitively demonstrates the impact that is generated by the position jitters of UAVs. Figs. 11(a) and 11(b) illustrate that the minimum receiving SNR of BSs is diminishing and the maximum average receiving SNR of AUs is increasing with the jitters of UAVs. The reason is that the beam pattern of the UVAA system is deteriorated. Moreover, Fig. 11(c) shows that the propulsion power consumption of UAVs will also be influenced, and it tends to be larger with the distance of jitter becomes further. However, the minimum receiving SNR of BSs is still above 2 dB even when the maximum jitter range of UAVs reaches 1 m, which corresponds that the achievable rate is above 2.32×10^6 bps. In conclusion, the performance of the UVAA system will be slightly influenced if the position jitters occur.

3) *Position Jitters of AUs*: The AUs will also encounter the position jitters, thus we also conduct simulations to test whether the maximum average receiving SNR of AUs will change significantly when the jitters occur. Specifically, the maximum jitters of the AUs in 3D directions are the same as VI-C.2. Note that the receiving SNR of BSs and the propulsion power consumption of UAVs are not considered since they will not be influenced by the position jitters of AUs.

It is found from Fig. 12 that the maximum average receiving SNR of AUs is increasing with the jitters of UAVs enlarge. The reason is that there will be other sidelobes pointing at

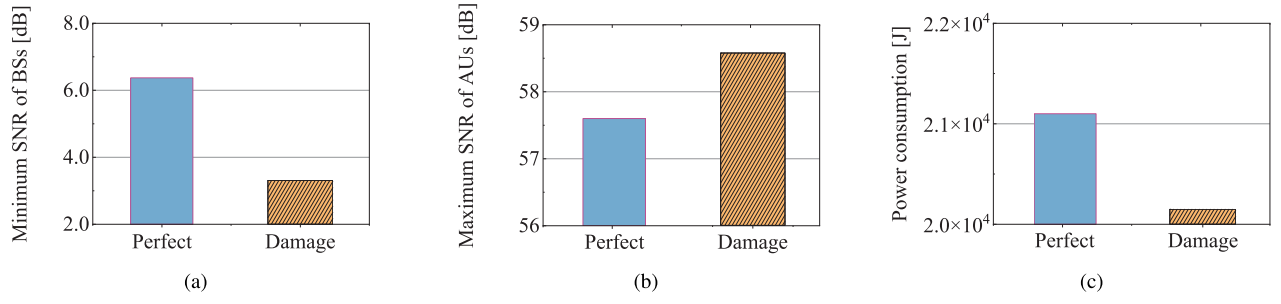


Fig. 10. Performance analysis of the UVAA system influenced by a damaged UAV. (a) Minimum SNR of BSs with a damaged UAV. (b) Maximum SNR of AUs with a damaged UAV. (c) Propulsion power consumption of UAVs with a damaged UAV.

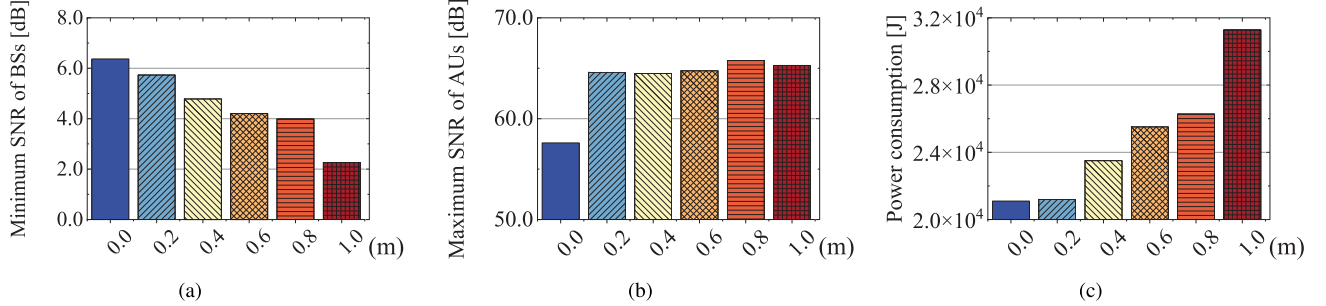


Fig. 11. Performance analysis of the UVAA system influenced by position jitters of UAVs. (a) Minimum SNR of BSs with position jitters. (b) Maximum SNR of AUs with position jitters. (c) Propulsion power consumption of UAVs with position jitters.

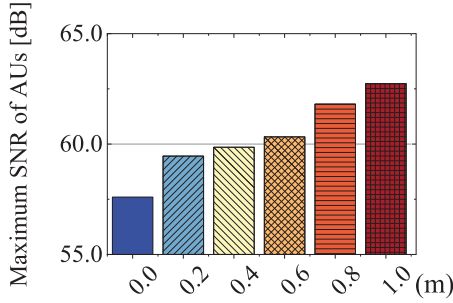


Fig. 12. Maximum receiving SNR of AUs influenced by position jitters of AUs.

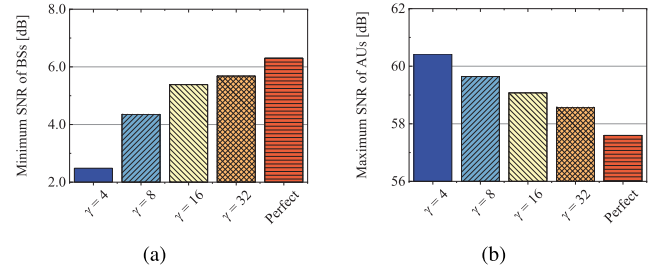


Fig. 13. Performance analysis of the UVAA system influenced by imperfect phase synchronization. (a) Minimum SNR of BSs with phase errors. (b) Maximum SNR of AUs with phase errors.

these AUs once upon the position jitters happen, while these sidelobes are not optimized and the corresponding radiation intensity is relatively high, resulting in the increase of the maximum average receiving SNR of AUs. To suppress the maximum SLL can overcome the abovementioned challenge brought by the position jitters of AUs, which will be left as our future work.

4) *Imperfect Phase Synchronization*: In this part, simulations are carried out to present the impact brought by the phase errors among UAVs on the performance of the UVAA system. Accordingly, the AF of the UVAA system is redefined as follows [75]:

$$F^{\zeta}(\theta, \phi) = \sum_{i=1}^{N_U} I_i e^{i_u[k_c(x_i^U \sin \theta \cos \phi + y_i^U \sin \theta \sin \phi + z_i^U \cos \theta) + \zeta_i]}, \quad (36)$$

where ζ_i represents the phase error of the i th UAV and it is assumed to obey the Tikhonov distribution with parameter γ which determines the error size [76]. Note that the propulsion

power consumption of UAVs is not considered in this part since it is only related with the locations of UAVs.

Fig. 13 shows the obtained results through considering the phase errors. Specifically, Fig. 13(a) shows that the minimum receiving SNR of BSs will be enhanced with the increasing of γ , and Fig. 13(b) manifests the maximum average receiving SNR of AUs tends to diminish with the γ increases. Generally speaking, the imperfect phase synchronization inevitably has a mild influence on the performance of the UVAA system, however, the impact is gradually overcome with the propose of synchronization algorithms.

VII. CONCLUSION

In this work, the UVAA-enabled emergency communication scenario was investigated. Specifically, we considered a UAV-based relay system that harvests data from ground users and then sends the collected data to several BSs via using CB. We formulated an RECMOP to cooperatively maximize the minimum receiving SNR of the BSs, minimize

the maximum average receiving SNR of the neighbouring AUs and minimize the propulsion power consumption of the UAVs. To deal with the formulated RECMOP which is shown to be NP-hard and non-convex, an IMOCSA was proposed. Simulations were implemented and the results manifested the effectiveness of the proposed IMOCSA. Moreover, the results also illustrated that the proposed algorithm outperforms other benchmark schemes. In addition, several unexpected circumstances were analyzed and discussed, and the results suggested that the proposed CB-based approach will inevitably be affected by some exceptional cases, however, it is still able to complete data transmission. In the future, we will extend the proposed CB-based approach to more UAV-assisted wireless communication scenarios including UAVs-assisted wireless communications in post-disaster areas, wireless communications from UAVs to satellites, and UAVs-enabled covert communications.

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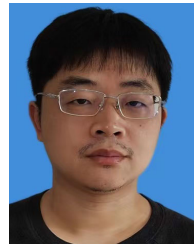
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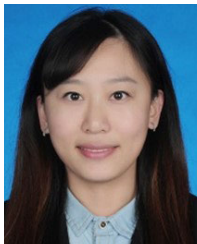
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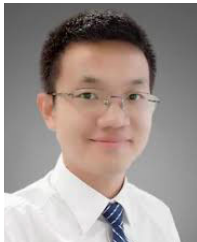
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